22. Edited version of K. Hentzelt: On the Theory of Polynomial Ideals and Resultants (continued)

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§ 4. Connection between ideals of polynomials and modules of linear forms.

To regard an ideal $\mathfrak{m}$ of polynomials, which is always assumed transformed, as a module $\mathfrak{M}_{i-1}$ $(i = 1, \ldots, n)$ of linear forms, the individual polynomials $F(x)$ are to be considered as linear forms in the power products ξ_λ of $x_1 \ldots x_{i-1}$:

$$F(x) = \sum a_\lambda^{(i)} \xi_\lambda,$$

where, according to the notation of § 3, the $a_\lambda^{(i)}$ are polynomials in $x_i \ldots x_n$. From the definition of the ideal it follows at once that $\mathfrak{M}_{i-1}$ has the module property with respect to these integral quantities $a^{(i)}, b^{(i)}, \ldots$; and since the infinitely many power products ξ of $x_1 \ldots x_{i-1}$ occur only linearly, by reason of their linear independence they play for $\mathfrak{M}_{i-1}$ the role of indeterminates. Each module $\mathfrak{M}_{i-1}$ contains infinitely many indeterminates ξ, but in each individual linear form only finitely many indeterminates occur. Likewise each fundamental ideal $\mathfrak{g}_{i-1}$ is to be regarded as a module $\mathfrak{G}_{i-1}$ of linear forms, where the indeterminates are again the power products of $x_1 \ldots x_{i-1}$. For $i = 1$ there is only one indeterminate ξ_0, corresponding to the unit.

One can now, and this is the basis of everything that follows, restrict oneself, despite these infinitely many indeterminates ξ, to modules of linear forms in finitely many indeterminates, according to the following theorem.

Theorem VII. *The residue-class system $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ is isomorphic to a residue-class system $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$, where $\mathfrak{M}^*_{i-1}$ is a module in finitely many indeterminates and $\mathfrak{G}^*_{i-1}$ is its fundamental module. The module $\mathfrak{M}_{i-1}$ has – after adjoining $x_{i+1} \ldots x_n$ to $P(u)$ – only finitely many elementary divisors different from 1, namely the elementary divisors of $\mathfrak{M}^*_{i-1}$ that are different from 1.*

The elementary divisors of $\mathfrak{M}_{i-1}$ are to be defined as in note 8). The isomorphism occurring in Theorem VII rests on the following definition.

Definition V. *Two residue-class systems are called isomorphic if they can be put in one-to-one correspondence in such a way that the difference of two classes is assigned to the difference of the corresponding classes, and the product of a class by an integral quantity to the product of the corresponding class by the same integral quantity.*

The proof of Theorem VII is obtained by complete induction. For $i = 1$ the theorem is evident: $\mathfrak{M}_0$ is a module of linear forms in one indeterminate ξ_0, corresponding to the power product of dimension zero of the x, that is, to the unit; the integral quantities are all polynomials in $x_1 \ldots x_n$. The fundamental ideal $\mathfrak{g}_0$ is the unit ideal consisting of all polynomials in $x_1 \ldots x_n$, and hence $\mathfrak{G}_0$ is the module (ξ_0), the fundamental module of $\mathfrak{M}_0$; thus here $\mathfrak{G}^*_0 \mid \mathfrak{M}^*_0$ coincides with $\mathfrak{G}_0 \mid \mathfrak{M}_0$. Finally, since $\mathfrak{M}_0$ has rank 1, it can have at most one elementary divisor different from 1.

We first pass from $i = 1$ to $i = 2$, in order to use the insight thereby obtained into the structure of $\mathfrak{G}_1 \mid \mathfrak{M}_1$ for the general induction step. For this purpose we first show that for $\mathfrak{G}_0 \mid \mathfrak{M}_0$ an x_1-bounded system of representatives may be chosen; because of the divisibility of $\mathfrak{g}_1$ by $\mathfrak{g}_0$, this will then lead to the finitely many indeterminates of $\mathfrak{G}^*_1$.

As a transformed ideal, $\mathfrak{m}$ possesses by § 3 at least one polynomial $C^{(1)}(x)$ regular in x_1, say of dimension k; hence $\mathfrak{M}_0$ contains the linear form $C^{(1)}\xi_0$. By the regularity of $C^{(1)}(x)$, every polynomial $H(x)$ admits a representation

$$H(x) = b_0^{(2)} + b_1^{(2)} x_1 + \cdots + b_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)); \tag{21}$$

therefore, since $C^{(1)}\xi_0$ belongs to $\mathfrak{M}_0$, one can indeed choose for $\mathfrak{G}_0 \mid \mathfrak{M}_0$ a system of representatives that does not reach the k-th dimension in x_1.

If now, in particular, for the polynomials $G(x)$ from $\mathfrak{g}_1$ and $F(x)$ from $\mathfrak{m}$ one writes

$$\begin{cases} G(x) = g_0^{(2)} + g_1^{(2)} x_1 + \cdots + g_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)), \\ F(x) = a_0^{(2)} + a_1^{(2)} x_1 + \cdots + a_{k-1}^{(2)} x_1^{k-1} \quad (C^{(1)}(x)), \end{cases} \tag{22}$$

and if $\mathfrak{G}_1^*$ denotes the system of all linear forms $g(\xi) = g_0^{(2)}\xi_0 + \cdots + g_{k-1}^{(2)}\xi_{k-1}$, and correspondingly $\mathfrak{M}_1^*$ the system of the linear forms $a(\xi) = a_0^{(2)}\xi_0 + \cdots + a_{k-1}^{(2)}\xi_{k-1}$, then the ideal property of $\mathfrak{g}_1$ and $\mathfrak{m}$ shows that $\mathfrak{G}_1^*$ and $\mathfrak{M}_1^*$ are modules of linear forms in $\xi_0 \dots \xi_{k-1}$ with respect to the integral quantities $b^{(2)}, c^{(2)}, \dots$. Likewise the principal ideal derived from $C^{(1)}(x)$ gives a module with respect to these integral quantities,

$$\mathfrak{G}_1 = (\xi_0, \xi_1, \dots, \xi_r, \dots),$$

where $\xi_r = x_1^r C^{(1)}(x)$. The ξ_r are linearly independent with respect to these integral quantities $b^{(2)}, c^{(2)}, \dots$, both among themselves and from the finitely many ξ. From the divisibility of $\mathfrak{G}_1$ by $\mathfrak{M}_1$, and hence by $\mathfrak{G}_1$, it follows that $\mathfrak{G}_1^*$ is divisible by $\mathfrak{G}_1$ and $\mathfrak{M}_1^*$ by $\mathfrak{M}_1$. Taking (22) into account, one obtains

$$\mathfrak{G}_1 = (\mathfrak{G}_1^*, \mathfrak{G}_1), \qquad \mathfrak{M}_1 = (\mathfrak{M}_1^*, \mathfrak{G}_1). \tag{23}$$

From (23) and from the linear independence of the ξ and ξ, Theorem VII for $i = 2$ follows as follows. Because $\mathfrak{G}_1$ is divisible by $\mathfrak{M}_1$, one first has $\mathfrak{G}_1 \mid \mathfrak{M}_1 = \mathfrak{G}_1^* \mid \mathfrak{M}_1$. To prove the isomorphism with $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$, let certain linear forms $g^*(\xi)$ from $\mathfrak{G}_1^*$ give a system of representatives of $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$. Because of the linear independence of the ξ and ξ, from $g^*(\xi) \equiv 0(\mathfrak{M}_1)$ it follows also that $g^*(\xi) \equiv 0(\mathfrak{M}_1^*)$; the $g^*(\xi)$ are therefore also incongruent modulo $\mathfrak{M}_1$, form a system of representatives of $\mathfrak{G}_1 \mid \mathfrak{M}_1$, and the correspondence from $\mathfrak{G}_1 \mid \mathfrak{M}_1$ to $\mathfrak{G}_1^* \mid \mathfrak{M}_1^*$ mediated by this system of representatives is isomorphic in the sense of Definition V.

In exactly the same way one obtains: from $b^{(2)}g(\xi) \equiv 0(\mathfrak{M}_1)$, $b^{(2)} \neq 0$, follows $b^{(2)}g(\xi) \equiv 0(\mathfrak{M}_1^*)$, $b^{(2)} \neq 0$. Since this is satisfied for all $g(\xi)$ from $\mathfrak{G}_1^*$, and only for these – for by (23) $\mathfrak{G}_1^*$ consists of all polynomials of the fundamental ideal $\mathfrak{g}_1$ that are at the same time linear forms in ξ – $\mathfrak{G}_1^*$ is thereby characterized as the fundamental module of $\mathfrak{M}_1^*$.

Adjoining finally $x_3 \dots x_n$ to $P(u)$, one gets

$$\begin{cases} \mathfrak{G}_1^* = (\eta_1, \dots, \eta_\rho), & \mathfrak{M}_1^* = (e_1\eta_1, \dots, e_\rho\eta_\rho), \\ \mathfrak{G}_1 = (\eta_\rho, \dots, \eta_1, \xi_0, \dots, \xi_r, \dots), & \mathfrak{M}_1 = (e_\rho\eta_\rho, \dots, e_1\eta_1, \xi_0, \dots, \xi_r, \dots), \end{cases} \tag{24}$$

and hence

$$(e_\rho) = \mathfrak{M}_1/\mathfrak{G}_1 = \mathfrak{M}_1/(\mathfrak{M}_1, \eta_\rho), \qquad (e_{\rho-1}) = (\mathfrak{M}_1, \eta_\rho)/\mathfrak{G}_1 \quad \text{etc.}$$

Thus, taking note 8) into account, $e_\rho, e_{\rho-1}, \dots, e_1, 1, \dots, 1, \dots$ are recognized as the elementary divisors of $\mathfrak{M}_1$. This proves all parts of Theorem VII for $i = 2$.

It should be noted that (22) and (23) use only that $C^{(1)}(x)$ is regular in x_1. These formulae, and the arguments attached to them leading to Theorem VII, therefore remain valid if, in place of (12), the special transformation

$$y_1 = x_1; \quad y_2 = u_{21}x_1 + z_2; \quad \dots; \quad y_n = u_{n1}x_1 + z_n \tag{25}$$

is taken as the basis; by composition with

$$\begin{cases} z = U_1(x) \quad \text{or} \\ z_2 = x_2; \quad z_3 = u_{32}x_2 + x_3; \quad \dots; \quad z_n = u_{n2}x_2 + \cdots + u_{n,n-1}x_{n-1} + x_n, \end{cases} \tag{26}$$

this again gives (12). If $\widetilde{\mathfrak{m}}_{(u)}$ passes by means of (25) into $\widetilde{\mathfrak{m}}$, and if $\widetilde{\mathfrak{g}}_1, \widetilde{\mathfrak{G}}_1, \dots$ have for $\widetilde{\mathfrak{m}}$ the same meaning as $\mathfrak{g}_1, \mathfrak{G}_1, \dots$ have for $\mathfrak{m}$, then

$$\widetilde{\mathfrak{G}}_1 = (\widetilde{\mathfrak{G}}_1^*, \widetilde{\mathfrak{G}}_1), \qquad \widetilde{\mathfrak{M}}_1 = (\widetilde{\mathfrak{M}}_1^*, \widetilde{\mathfrak{G}}_1). \tag{27}$$

Here (27) arises from (23) simply by inversion of (26). This is evident for $\mathfrak{m}$ and $C^{(1)}(x)$, and hence also for $\mathfrak{G}_1$ and $\mathfrak{M}_1$. It is also true for $\mathfrak{g}_1$, since

$$b^{(2)}(x)G(x) \equiv 0(\mathfrak{m}), \quad b^{(2)} \neq 0, \qquad \widetilde{b}^{(2)}(z)\widetilde{G}(x,z) \equiv 0(\widetilde{\mathfrak{m}}), \quad \widetilde{b}^{(2)} \neq 0$$

mutually condition one another through $z = U_1(x)$. Thus $\widetilde{\mathfrak{g}}_1 = \mathfrak{g}_1$ by means of (26); hence also $\widetilde{\mathfrak{G}}_1 = \mathfrak{G}_1$. Consequently, by the definition given through (22), the same holds for $\mathfrak{G}_1^*$ and $\mathfrak{M}_1^*$ as for $\widetilde{\mathfrak{G}}_1^*$ and $\widetilde{\mathfrak{M}}_1^*$, and so for the whole representation (27).

For the general induction step, suppose the following hypotheses hold:

$$\begin{cases} \mathfrak{G}_{i-1} = (\mathfrak{G}_{i-1}^*, \mathfrak{G}_{i-1}), & \mathfrak{M}_{i-1} = (\mathfrak{M}_{i-1}^*, \mathfrak{G}_{i-1}), \\ \mathfrak{G}_{i-1} = (\xi_0, \xi_1, \dots, \xi_r, \dots), \end{cases} \tag{28}$$

where $\mathfrak{G}^*_{i-1}$ and $\mathfrak{M}^*_{i-1}$ are modules, with respect to the integral quantities $b^{(i)}, c^{(i)}, \dots$, of linear forms in finitely many indeterminates ξ, certain power products of $x_1, \dots, x_{i-1}$; and where the ξ are linearly independent, both among themselves and from the ξ, with respect to these integral quantities. A representation corresponding to (28), and the linear independence of the ξ and ξ, is also to hold when, instead of (12), one uses the special transformation

$$\begin{aligned} y_1 = x_1; \quad \dots; \quad y_{i-1} = u_{i-1,1}x_1 + \dots + x_{i-1}; \\ y_i = u_{i,1}x_1 + \dots + u_{i,i-1}x_{i-1} + z_i; \quad \dots; \quad y_n = u_{n,1}x_1 + \dots + u_{n,i-1}x_{i-1} + z_n, \end{aligned} \tag{29}$$

which can be composed to give (12) by means of

$$z = U_{i-1}(x) \quad \text{or} \quad z_i = x_i; \quad z_{i+1} = u_{i+1,i}x_i + x_{i+1}; \quad \dots; \quad z_n = u_{n,i}x_i + \dots + u_{n,n-1}x_{n-1} + x_n.$$

Moreover this special representation is to arise from (28) simply by inversion of $z = U_{i-1}(x)$.

From the hypotheses (28) and the linear independence of the ξ and ξ, the isomorphism of $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ with $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$ follows by assignment to the same system of representatives, by exactly the same arguments as those attached above to (23). Likewise it follows that $\mathfrak{G}^*_{i-1}$ becomes the fundamental module of $\mathfrak{M}^*_{i-1}$, and that $\mathfrak{M}_{i-1}$ has only finitely many elementary divisors different from 1, namely those elementary divisors of $\mathfrak{M}^*_{i-1}$ that are different from 1.

To carry out the induction step it remains first to show again that for $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$, and hence also for $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$, an x_i-bounded system of representatives may be chosen. This follows from the quotient representation proved in (7), Theorem III:

$$\mathfrak{G}^*_{i-1} = \mathfrak{M}^*_{i-1}/(C^{(i)}), \tag{30}$$

where $C^{(i)}$ means any non-identically vanishing determinant of degree ρ from $\mathfrak{M}^*_{i-1}$, ρ being the rank of $\mathfrak{M}^*_{i-1}$. From the part of the hypotheses referring to the special transformation (29), it follows that $C^{(i)}$ may always be assumed regular with respect to x_i. For the module $\widetilde{\mathfrak{M}}^*_{i-1}$ corresponding to $\mathfrak{M}^*_{i-1}$ has the same rank; if $\widetilde{C}^{(i)}$ denotes any non-identically vanishing determinant of degree ρ from $\widetilde{\mathfrak{M}}^*_{i-1}$ which passes by $z = U_{i-1}(x)$ into a regular $C^{(i)}$, then by hypothesis $C^{(i)}$ is a determinant of degree ρ from $\mathfrak{M}^*_{i-1}$.

From the existence of $C^{(i)}$ it follows, as in (8), with a suitable numbering of the ξ, that ρ linear forms of the special shape

$$L_\lambda(\xi) = C^{(i)}\xi_\lambda + c^{(i)}_{\lambda,1}\xi_{\rho+1} + \dots + c^{(i)}_{\lambda,s-\rho}\xi_s \qquad (\lambda = 1 \dots \rho)$$

belong to $\mathfrak{M}^*_{i-1}$. Every linear form in ξ can now be brought into the form

$$H(\xi) = a^{(i)}_1\xi_1 + \dots + a^{(i)}_\rho\xi_\rho + b^{(i)}_1\xi_{\rho+1} + \dots + b^{(i)}_{s-\rho}\xi_s \quad (L_1(\xi), \dots, L_\rho(\xi)), \tag{31}$$

where $a^{(i)}_1 \dots a^{(i)}_\rho$ are bounded in x_i and do not reach the degree k of $C^{(i)}$. This representation is unique; for from

$$a^{(i)}_1\xi_1 + \dots + a^{(i)}_\rho\xi_\rho + b^{(i)}_1\xi_{\rho+1} + \dots + b^{(i)}_{s-\rho}\xi_s \equiv 0 \quad (L_1(\xi), \dots, L_\rho(\xi))$$

it follows, by comparing coefficients in $\xi_1 \dots \xi_\rho$ and considering the degrees in x_i, that all $a^{(i)}$ and $b^{(i)}$ vanish identically.

Let now in particular

$$H(\xi) \equiv 0(\mathfrak{G}^*_{i-1}); \qquad \text{hence} \qquad C^{(i)}H(\xi) \equiv 0(\mathfrak{M}^*_{i-1}) \quad \text{by (30).}$$

As in the proof of Theorem III one obtains

$$C^{(i)}H(\xi) - \sum_{\tau=1}^{s-\rho}\{C^{(i)}b^{(i)}_\tau - \sum_\lambda a^{(i)}_\lambda c^{(i)}_{\lambda,\tau}\}\xi_{\rho+\tau} \equiv 0(\mathfrak{M}^*_{i-1}),$$

and therefore, since as in Theorem III the coefficients of $\xi_{\rho+1} \dots \xi_s$ must vanish,

$$C^{(i)}b^{(i)}_\tau = \sum_\lambda a^{(i)}_\lambda c^{(i)}_{\lambda,\tau} \qquad (\tau = 1 \dots s-\rho).$$

Thus the $b^{(i)}_\tau$ are also bounded and do not reach the maximal degree of the $c^{(i)}_{\lambda,\tau}$. The existence of an x_i-bounded system of representatives for $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$, that is for $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$, not reaching a certain fixed degree r, is thereby proved.

Let $\vartheta_1 \dots \vartheta_t$ denote the finitely many power products

$$x_i^\alpha \xi_\lambda \quad (\alpha = 0, 1, \dots, k-1;\ \lambda = 1, 2, \dots, \rho), \qquad x_i^\beta \xi_{\rho+\tau} \quad (\beta = 0, 1, \dots, r-1;\ \tau = 1, 2, \dots, s-\rho)$$

and arrange the countably infinite expressions

$$x_i^\gamma L_\lambda \quad (\gamma = 0, 1, \dots \text{ in inf.};\ \lambda = 1, 2, \dots, \rho), \qquad x_i^\gamma \xi_\nu \quad (\gamma, \nu = 0, 1, \dots \text{ in inf.})$$

in a simply infinite sequence $\omega_0, \omega_1, \dots, \omega_\mu, \dots$. Then the ϑ and ω are linearly independent, both each among themselves and from one another, with respect to the integral quantities $b^{(i+1)}, c^{(i+1)}, \dots$. For from

$$\sum c_\alpha^{(i+1)} x_i^\alpha \xi_\lambda + \sum c_\beta^{(i+1)} x_i^\beta \xi_{\rho+\tau} + \sum d_\gamma^{(i+1)} x_i^\gamma L_\lambda(\xi) + \sum e_{\gamma\nu}^{(i+1)} x_i^\gamma \xi_\nu = 0,$$

each sum extending over finitely many terms, the assumed linear independence of the ξ and ξ, and of the ξ among themselves with respect to the integral quantities $b^{(i)}$, gives the vanishing of all $e_{\gamma\nu}^{(i+1)}$. The uniqueness of the representation (31) further gives the vanishing of the $c_\alpha^{(i+1)}$ and $c_\beta^{(i+1)}$, and finally, because of the linear independence of the $L_\lambda(\xi)$, the vanishing of the $d_\gamma^{(i+1)}$.

If the module $(\mathfrak{G}_{i-1}, L_1(\xi), \dots, L_\rho(\xi))$ is now regarded as a module $\mathfrak{G}_i$ with respect to the integral quantities $b^{(i+1)}$, then $\mathfrak{G}_i$ is divisible by $\mathfrak{M}_i$, because $\mathfrak{G}_{i-1}, L_1(\xi), \dots, L_\rho(\xi)$ are divisible by $\mathfrak{M}_{i-1}$, and

$$\mathfrak{G}_i = (\omega_0, \omega_1, \dots, \omega_\mu, \dots).$$

For each $H(x) \equiv 0(\mathfrak{g}_{i-1})$, (28), (31), and what has just been proved give

$$H(x) = b_1^{(i+1)} \vartheta_1 + \dots + b_t^{(i+1)} \vartheta_t \quad (\mathfrak{G}_i)$$

as a module equation with respect to the integral quantities $b^{(i+1)}, c^{(i+1)}$. Since $\mathfrak{g}_i$ is divisible by $\mathfrak{g}_{i-1}$ and $\mathfrak{m}$ is divisible by $\mathfrak{g}_i$, if one now sets, in particular, for every $G(x)$ from $\mathfrak{G}_i$ and every $F(x)$ from $\mathfrak{M}_i$,

$$G(x) = g_1^{(i+1)} \vartheta_1 + \dots + g_t^{(i+1)} \vartheta_t \quad (\mathfrak{G}_i), \qquad F(x) = a_1^{(i+1)} \vartheta_1 + \dots + a_t^{(i+1)} \vartheta_t \quad (\mathfrak{G}_i),$$

and denotes by $\mathfrak{G}_i^*$ the module consisting of all $g(\vartheta)$, and by $\mathfrak{M}_i^*$ the module consisting of all $a(\vartheta)$, then exactly the arguments leading to (23) give

$$\mathfrak{G}_i = (\mathfrak{G}_i^*, \mathfrak{G}_i), \qquad \mathfrak{M}_i = (\mathfrak{M}_i^*, \mathfrak{G}_i), \qquad \mathfrak{G}_i = (\omega_0, \omega_1, \dots, \omega_\mu, \dots). \tag{32}$$

In deriving (32), again only the regularity of $C^{(i)}$ in x_i was used, so the whole argument remains valid if the special transformation (29) is taken with the index increased by one. The same considerations as those attached to (27) then show that this special representation arises from (32) simply by the inversion of $z = U_i(x)$.

Thus all hypotheses (28), etc., of the general induction step have been proved for the next index; and since, as shown there, Theorem VII follows directly from these hypotheses, Theorem VII is proved in general. At the same time it has been shown that the arbitrariness in $(\mathfrak{G}_{i-1}^*, \mathfrak{M}_{i-1}^*)$ caused by the arbitrary choice of $C^{(i)}$ is again removed by the isomorphism of the residue-class system $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$.

In particular, if the field P used in § 3 as coefficient domain has characteristic zero – that is, if the prime field contained in P and derived from the unit is of the type of the rational numbers – then the indeterminates $u_{\mu\nu}$ can be specialized to quantities $\bar{u}_{\mu\nu}$ from P in such a way that for $i = 1 \dots n$ each $C^{(i)}$ remains regular in x_i. The above arguments show that Theorem VII remains valid under this specialization as well.[1] The fundamental module and elementary divisors need not, however, arise necessarily from the general case by the specialization $u_{\mu\nu} = \bar{u}_{\mu\nu}$.[2]

[1] Hentzelt takes, instead of an arbitrary P, only the field of all complex numbers and treats chiefly this latter case, whereas he uses indeterminates only in the actual elimination theory. Hentzelt then further shows, essentially by the arguments given here, that in order to form the resultant form it suffices to go, in each $\mathfrak{M}_{i-1}$, only up to some finite degree in the x that exceeds a fixed bound. What is missing is the isomorphism of $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ with $\mathfrak{G}_{i-1}^* \mid \mathfrak{M}_{i-1}^*$ given in Theorem VII, which explains the independence from the degree numbers. (E. N.)

[2] For $\mathfrak{m} = (x^2, ux + y)$ one has $\mathfrak{g}_0 = (1)$, $\mathfrak{g}_1 = (1)$. If $C^{(1)} = x^2$ is chosen, then $\mathfrak{G}_1^* = (1, x)$, $\mathfrak{M}_1^* = (ux + y, xy)$; this $\mathfrak{M}_1^*$ has elementary divisors y^2 and 1, and $C^{(2)} = y^2$ can be chosen. For $u = 0$, $C^{(1)}$ and $C^{(2)}$ remain regular in x and y respectively; but $\mathfrak{M}_1^* = (y, xy)$ has elementary divisors y, y. Thus $\mathfrak{G}_1 \mid \mathfrak{M}_1$ is still isomorphic to the residue-class system of a finite module, but is not isomorphic to $\mathfrak{G}_1 \mid \mathfrak{M}_1$. If instead one had chosen $C^{(1)} = ux + y$ and specialized so that $C^{(1)}$ remained regular, the isomorphism would also have been preserved. (E. N.)

§ 5. Resultant form and elementary-divisor form of an ideal of polynomials.

Let $x_{i+1} \dots x_n$ be adjoined to $P(u)$, so that $\mathfrak{G}_{i-1}$ and $\mathfrak{M}_{i-1}$, and therefore also $\mathfrak{G}^*_{i-1}$ and $\mathfrak{M}^*_{i-1}$, pass into modules of linear forms in which the integral quantities can be regarded as polynomials in one indeterminate x_i. By Theorem VII, in this case the elementary divisors different from 1 of $\mathfrak{M}_{i-1}$, together with at most finitely many elementary divisors equal to 1, agree with those of $\mathfrak{M}^*_{i-1}$. The product of these elementary divisors, the ρ-th determinantal divisor of $\mathfrak{M}^*_{i-1}$, was equal to the determinant of the transition substitution from $\mathfrak{G}^*_{i-1}$ to $\mathfrak{M}^*_{i-1}$, hence to $N(\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1})$ by Definition III. Formula (24), or the analogous formula following from (28) for general i, shows that the product of these elementary divisors is also equal to the determinant of the transition substitution from $\mathfrak{G}_{i-1}$ to $\mathfrak{M}_{i-1}$, and is therefore to be denoted by $N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1})$. Thus

$$N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}) = N(\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}) = R^{(i)}(x),$$

where the polynomial $R^{(i)}(x)$, that is, the greatest common divisor, in the polynomial sense, of all ρ-rowed determinants from $\mathfrak{M}^*_{i-1}$, may be assumed integral and primitive in $x_{i+1} \dots x_n$, and consequently, as a divisor of $C^{(i)}$, becomes regular in x_i. Likewise the highest elementary divisor $E^{(i)}(x)$ of $\mathfrak{M}_{i-1}$, respectively of $\mathfrak{M}^*_{i-1}$, may be assumed integral and primitive in $x_{i+1} \dots x_n$ and hence regular in x_i. This leads to the following definition.

Definition VI. *The polynomial $R^{(i)}(x)$ is called the resultant of the i-th stage of $\mathfrak{m}$, and $E^{(i)}(x)$ the elementary divisor of the i-th stage. Thus the resultant of the i-th stage is the norm of the fundamental ideal $\mathfrak{g}_{i-1}$ with respect to $\mathfrak{m}$, interpreted as the norm of the module $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$, where $x_{i+1} \dots x_n$ are adjoined to $P(u)$; likewise $E^{(i)}(x)$, under the same adjunction, becomes equal to the quotient $\mathfrak{M}_{i-1}/\mathfrak{G}_{i-1}$. The resultant form and elementary-divisor form of $\mathfrak{m}$ are the products*

$$R_{\mathfrak{m}} = R^{(1)} R^{(2)} \cdots R^{(n)}, \qquad E_{\mathfrak{m}} = E^{(1)} E^{(2)} \cdots E^{(n)}.$$

For resultant form and elementary-divisor form one has:

Theorem VIII. *The resultant form is divisible by the elementary form; conversely, a power of $E_{\mathfrak{m}}$ is divisible by $R_{\mathfrak{m}}$. The forms $E_{\mathfrak{m}}$ and $R_{\mathfrak{m}}$ are divisible by $\mathfrak{m}$; more generally, $E^{(i)} \cdots E^{(n)} \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$ and consequently $R^{(i)} \cdots R^{(n)} \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$.*

Indeed, since the norm is divisible by the highest elementary divisor and conversely a power of this highest elementary divisor is divisible by the norm, this divisibility follows for $R^{(i)}(x)$ and $E^{(i)}(x)$ after adjoining $x_{i+1} \dots x_n$. But $R^{(i)}(x)$ and $E^{(i)}(x)$ are assumed primitive polynomials with respect to $x_{i+1} \dots x_n$; hence the divisibility holds with respect to all indeterminates $x_i, x_{i+1}, \dots, x_n$. Thus $R_{\mathfrak{m}}$ is divisible by $E_{\mathfrak{m}}$, and some power of $E_{\mathfrak{m}}$ by $R_{\mathfrak{m}}$, according to the definition of these expressions in all indeterminates x.

Furthermore, by Theorems VII and II, after adjoining $x_{i+1} \dots x_n$, the highest elementary divisor $E^{(i)}(x)$ is defined as the quotient $\mathfrak{M}_{i-1}/\mathfrak{G}_{i-1}$. Thus, returning to polynomials in all indeterminates x, for every

$$G(x) \equiv 0(\mathfrak{g}_{i-1}) \quad \text{there is a} \quad b^{(i+1)} \neq 0 \quad \text{such that} \quad b^{(i+1)} E^{(i)}(x) G(x) \equiv 0(\mathfrak{m}). \tag{33}$$

In particular $\mathfrak{g}_0$ is the unit ideal, whence

$$b^{(2)} E^{(1)}(x) \equiv 0(\mathfrak{m}), \quad b^{(2)} \neq 0, \qquad \text{and thus} \qquad E^{(1)}(x) \equiv 0(\mathfrak{g}_1).$$

In general, from (33), applied to

$$E^{(1)}(x) \cdots E^{(i-1)}(x) \equiv 0(\mathfrak{g}_{i-1}),$$

there exists $b^{(i+1)} \neq 0$ such that $b^{(i+1)} E^{(1)} \cdots E^{(i)} \equiv 0(\mathfrak{m})$; hence $E^{(1)} \cdots E^{(i)} \equiv 0(\mathfrak{g}_i)$. Since $\mathfrak{g}_n = \mathfrak{m}$, one obtains

$$E_{\mathfrak{m}} = E^{(1)} \cdots E^{(n)} \equiv 0(\mathfrak{m}) \quad \text{and consequently} \quad R_{\mathfrak{m}} = R^{(1)} \cdots R^{(n)} \equiv 0(\mathfrak{m}). \tag{34}$$

If in (33) $G(x)$ runs through the finitely many polynomials of an ideal basis of $\mathfrak{g}_{i-1}$, and if $c^{(i+1)}(x)$ denotes the product of the corresponding $b^{(i+1)}(x)$, then

$$c^{(i+1)} E^{(i)}(x) \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m}); \qquad c^{(i+1)} \neq 0, \quad \text{and hence} \quad E^{(i)}(x) \mathfrak{g}_{i-1} \equiv 0(\mathfrak{g}_i),$$

and from this, as above, by finite repetition,

$$E^{(n)}(x) \cdots E^{(i)}(x) \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m})$$

and consequently also

$$R^{(n)}(x)\cdots R^{(i)}(x)\mathfrak{g}_{i-1}\equiv 0(\mathfrak{m}).$$

This proves all parts of Theorem VIII.

Theorem VIII depends essentially on properties of the elementary-divisor form; the characteristic significance of the resultant form for $\mathfrak{m}$ is shown by

Theorem IX. *If $\mathfrak{n}$ is a divisor of $\mathfrak{m}$ – divisor understood in the ideal sense – and if the resultant forms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then the ideals $\mathfrak{n}$ and $\mathfrak{m}$ agree.*

Let $\mathfrak{g}_0\ldots\mathfrak{g}_{n-1},\mathfrak{g}_n=\mathfrak{m}$ and $\mathfrak{h}_0\ldots\mathfrak{h}_{n-1},\mathfrak{h}_n=\mathfrak{n}$ denote the fundamental ideals of $\mathfrak{m}$ and $\mathfrak{n}$. First $\mathfrak{g}_0$ and $\mathfrak{h}_0$ are both the unit ideal, hence $\mathfrak{g}_0=\mathfrak{h}_0$. Assume now that $\mathfrak{g}_{i-1}=\mathfrak{h}_{i-1}$, so that $\mathfrak{M}_{i-1}$ and $\mathfrak{N}_{i-1}$ have the same fundamental module $\mathfrak{G}_{i-1}$. The hypothesis $R_{\mathfrak{n}}=R_{\mathfrak{m}}$, after adjoining $x_{i+1}\ldots x_n$, may be written as

$$N(\mathfrak{G}_{i-1}\mid\mathfrak{N}_{i-1})=N(\mathfrak{G}_{i-1}\mid\mathfrak{M}_{i-1})\quad\text{or}\quad N(\mathfrak{G}^*_{i-1}\mid\mathfrak{N}^*_{i-1})=N(\mathfrak{G}^*_{i-1}\mid\mathfrak{M}^*_{i-1}).$$

Here, in accordance with (32), $\mathfrak{N}_{i-1}=(\mathfrak{N}^*_{i-1},\mathfrak{G}_{i-1})$, so that $\mathfrak{N}^*_{i-1}$ is likewise a module of linear forms in ξ, with $\mathfrak{G}^*_{i-1}$ as its fundamental module. Since $\mathfrak{M}_{i-1}$ is divisible by $\mathfrak{N}_{i-1}$, which entails the divisibility of $\mathfrak{M}^*_{i-1}$ by $\mathfrak{N}^*_{i-1}$, Theorem IV shows that, under this adjunction, the two modules $\mathfrak{M}^*_{i-1}$ and $\mathfrak{N}^*_{i-1}$ agree, and consequently so do $\mathfrak{N}_{i-1}$ and $\mathfrak{M}_{i-1}$. But adjoining $x_{i+1}\ldots x_n$ means that one adds to $\mathfrak{M}_{i-1}$, respectively to $\mathfrak{m}$, all polynomials $G(x)$ for which

$$b^{(i+1)}(x)G(x)\equiv 0(\mathfrak{m}),\qquad b^{(i+1)}\neq 0.$$

Thus, by the adjunction, $\mathfrak{M}_{i-1}$, respectively $\mathfrak{m}$, passes into $\mathfrak{g}_i$, and correspondingly $\mathfrak{n}$ into $\mathfrak{h}_i$; therefore $\mathfrak{g}_i=\mathfrak{h}_i$. Hence finally $\mathfrak{g}_n=\mathfrak{h}_n$, i.e. $\mathfrak{m}=\mathfrak{n}$.

Finally the same transfer principle, applied to Theorem V, gives the following.

Theorem X. *Let $R^{(i)}=S^{(i)}T^{(i)}$ be a decomposition of $R^{(i)}$ into relatively prime polynomials $S^{(i)}$ and $T^{(i)}$ in the polynomial sense. Then there exists one and only one ideal $\mathfrak{j}$, and likewise one and only one ideal $\mathfrak{t}$, both divisors of $\mathfrak{m}$ with the same fundamental ideal of the $(i-1)$-st stage, $\mathfrak{g}_{i-1}$, such that $S^{(i)}=N(\mathfrak{G}_{i-1}\mid\mathfrak{S}_{i-1})$ and $T^{(i)}=N(\mathfrak{G}_{i-1}\mid\mathfrak{T}_{i-1})$. The ideals $\mathfrak{j}$ and $\mathfrak{t}$ are defined as the totality of polynomials $S(x)$ and $T(x)$, respectively, such that*

$$s^{(i+1)}(x)T^{(i)}(x)S(x)\equiv 0(\mathfrak{m}),\quad s^{(i+1)}\neq 0,$$
$$t^{(i+1)}(x)S^{(i)}(x)T(x)\equiv 0(\mathfrak{m}),\quad t^{(i+1)}\neq 0,$$

and $\mathfrak{g}_i$ becomes the least common multiple of $\mathfrak{j}$ and $\mathfrak{t}$,

$$\mathfrak{g}_i=[\mathfrak{j},\mathfrak{t}].$$

It remains only to note that $s^{(i+1)}$ and $t^{(i+1)}$ can be chosen fixed for $\mathfrak{j}$ and $\mathfrak{t}$, as the product of the $s^{(i+1)}$ and $t^{(i+1)}$ corresponding to the basis elements of $\mathfrak{j}$ and $\mathfrak{t}$, and that, as remarked above, by adjoining $x_{i+1}\ldots x_n$ the ideal $\mathfrak{m}$ is transformed into $\mathfrak{g}_i$. In particular, the decomposition of $R^{(i)}$ can be continued up to powers of irreducible polynomials – primary factors – which entails a corresponding representation of $\mathfrak{g}_i$ as a least common multiple.

§ 6. Proper elimination theory.

By Theorem VIII, respectively formula (34), the resultant form and the elementary-divisor form are divisible by $\mathfrak{m}$, and therefore vanish at all zeros of $\mathfrak{m}$. Here by "zeros of $\mathfrak{m}$" are meant value systems of $x_1\ldots x_i$ belonging to the algebraically closed field derived from $P(u;x_{i+1}\ldots x_n)$[3], such that every polynomial from $\mathfrak{m}$ vanishes at these value systems, while $x_{i+1}\ldots x_n$ are thought of as adjoined to the coefficient field $(i=1,\ldots,n)$. These adjoined $x_{i+1}\ldots x_n$ can, as will be shown, also be replaced by quantities from $P(u)$. The content of elimination theory, conversely, is the derivation of the zeros of $\mathfrak{m}$ from those of the resultant form. This converse rests on the successive elimination to be developed in this section; since in the next section the divisibility of the form $D^{(i)}(x)$ occurring in it by $E^{(i)}(x)$ will be proved, the desired converse follows from the divisibility of a power of $E^{(i)}(x)$ by $R^{(i)}(x)$.

[3]Steinitz showed, in his Algebraische Theorie der Körper (J. f. M. 137 (1910), pp. 167–309), that every field can be extended, essentially uniquely, to an algebraically closed one, thus giving the rational equivalent of the fundamental theorem of algebra. In fact, for each $i=1,\ldots,n$ one is dealing with a finite extension field of $P(u,x_{i+1},\ldots,x_n)$. (E. N.)

The theory of successive elimination to be given here rests on the following result.

Theorem XI. *Let $\mathfrak{b}$ be an ideal of polynomials in one indeterminate t with coefficients in P, hence a principal ideal; let $h(t)$ be a polynomial from $\mathfrak{b}$ of degree k, and let $\mathfrak{B}$ be the module of linear forms in $1, t, \ldots, t^{k-1}$ into which $\mathfrak{b}$ passes modulo $h(t)$. Then $\mathfrak{B}$ has rank $k-p$, where p is the degree of the basis polynomial $f(t)$ of $\mathfrak{b}$.*

By hypothesis $h(t) = h_1(t)f(t)$, where $h_1(t)$ has degree $k-p$. The residue classes of $\mathfrak{b}$ modulo $h(t)$ represented by $f(t), tf(t), \ldots, t^{k-p-1}f(t)$ are therefore linearly independent; and every residue class of $\mathfrak{b}$ modulo $h(t)$ can be represented linearly by these $k-p$ special classes, with coefficients in P. But modulo $h(t)$ the ideal $\mathfrak{b}$ passes into a module of linear forms in $1, t, \ldots, t^{k-1}$, whose rank is therefore exactly $k-p$.

Now let $\mathfrak{a} = \mathfrak{a}_1$ be a transformed ideal of polynomials, and let $A^{(1)}(x)$ be a polynomial from $\mathfrak{a}_1$, regular in x_1, of degree k_1. Let $\mathfrak{A}_1$ be the module of linear forms in $1, x_1, \ldots, x_1^{k_1-1}$, with polynomials $a^{(2)}(x)$ as coefficients, into which $\mathfrak{a}_1$ passes modulo $A^{(1)}(x)$. Let $\mathfrak{a}_2$ denote the ideal in $x_2 \ldots x_n$ derived from all k_1-rowed determinants from $\mathfrak{A}_1$. By Theorem XI, $\mathfrak{a}_2$ is the zero ideal if and only if the polynomials from $\mathfrak{a}_1$ have a common divisor, in the polynomial sense, of degree $p_1 > 0$ in x_1. If $\mathfrak{a}_2$ is different from the zero ideal, then, since $\mathfrak{a}$ is assumed transformed, it contains a polynomial $A^{(2)}(x)$ regular in x_2, of degree k_2, as is seen directly by composing the transformation (12) from the special transformations (25) and (26). Let $\mathfrak{A}_2$ again denote the module of linear forms in $1, x_2, \ldots, x_2^{k_2-1}$ into which $\mathfrak{a}_2$ passes modulo $A^{(2)}(x)$, and $\mathfrak{a}_3$ the ideal in $x_3 \ldots x_n$ derived from all k_2-rowed determinants from $\mathfrak{A}_2$; this ideal must contain a polynomial $A^{(3)}(x)$ regular in x_3.

In this way one continues the process until one reaches a zero ideal, or until $\mathfrak{a}_{n+1}$ becomes the unit ideal; for since $\mathfrak{a}_{n+1}$ is free of the x, it can only be the zero ideal or the unit ideal. It is seen that the ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots$ are uniquely determined by $\mathfrak{a}$, independently of the chosen polynomials $A^{(\lambda)}(x)$. This is clear for $\mathfrak{a}_1 = \mathfrak{a}$, and can therefore be assumed for $\mathfrak{a}_1, \ldots, \mathfrak{a}_i$. If $\mathfrak{a}_i$ passes modulo $A^{(i)}(x)$ into the module of linear forms $\mathfrak{A}_i$, then $\mathfrak{A}_i$ can also be characterized as the totality of the polynomials from $\mathfrak{a}_i$ that do not reach degree k_i in x_i; thus $\mathfrak{A}_i$ depends only on the degree k_i of $A^{(i)}(x)$. Let $\bar{A}^{(i)}(x)$ be a polynomial from $\mathfrak{a}_i$, regular in x_i, of degree $\bar{k}_i > k_i$, let $\overline{\mathfrak{A}}_i$ be the corresponding module of linear forms, and let $\bar{\mathfrak{a}}_{i+1}$ be the ideal derived from the $\bar{k}_i$-rowed determinants from $\overline{\mathfrak{A}}_i$. Then the module equation holds:

$$\overline{\mathfrak{A}}_i = (\mathfrak{A}_i, A^{(i)}, x_i A^{(i)}, \ldots, x_i^{\bar{k}_i - k_i - 1} A^{(i)}).$$

Since, by the regularity of $A^{(i)}(x)$ in x_i, the polynomials $A^{(i)}, x_i A^{(i)}, \ldots$ can be introduced as new indeterminates of the linear forms, it follows that the k_i-rowed determinants from $\mathfrak{A}_i$ agree with the $\bar{k}_i$-rowed determinants from $\overline{\mathfrak{A}}_i$, and therefore $\bar{\mathfrak{a}}_{i+1} = \mathfrak{a}_{i+1}$.[4]

Furthermore $\mathfrak{a}_{i+1}$ is always divisible by $\mathfrak{a}_i$. This is clear if $\mathfrak{a}_{i+1}$ is the zero ideal; in the opposite case $\mathfrak{A}_i$ has rank k_i, so that $\mathfrak{a}_{i+1}$ is the ideal which by definition consists of the k_i-rowed determinants from $\mathfrak{A}_i$, and is divisible by $\mathfrak{A}_i$ and hence by $\mathfrak{a}_i$. Thus every $\mathfrak{a}_{i+1}$ is divisible by $\mathfrak{a}$; if therefore $\mathfrak{a}_{n+1}$ becomes the unit ideal, then $\mathfrak{a}$ itself is the unit ideal.

Let now $\mathfrak{a}$ be different from the unit ideal, with $\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_i \neq 0$, and $\mathfrak{a}_{i+1} = 0$. Let $D^{(i)}(x)$ be the greatest common divisor, existing by Theorem XI, of all polynomials from $\mathfrak{a}_i$, where divisor is understood in the polynomial sense; as a divisor of $A^{(i)}(x)$, $D^{(i)}(x)$ is itself regular in x_i of degree $p_i > 0$. Adjoin $x_{i+1} \ldots x_n$ to $P(u)$; then $D^{(i)}(x)$ has, in an algebraic extension field of $P(u; x_{i+1} \ldots x_n)$, a decomposition into p_i linear factors. Let $x_i - \bar{x}_i$ be one such linear factor, so that all polynomials from $\mathfrak{a}_i$ vanish for $x_i = \bar{x}_i$; then, by Theorem XI, the polynomials from $\mathfrak{a}_{i-1}$ acquire for this specialization a greatest common divisor $D^{(i-1)}(x)$. As a divisor of $A^{(i-1)}(x)$, $D^{(i-1)}(x)$ is again regular in x_{i-1} of degree $p_{i-1} > 0$; in particular it cannot vanish identically, and in an extension field of $P(u, \bar{x}_i, x_{i+1}, \ldots, x_n)$ it decomposes into p_{i-1} linear factors. If $x_{i-1} - \bar{x}_{i-1}$ is such a linear factor, there corresponds to it a greatest common divisor from $\mathfrak{a}_{i-2}$. Continuing in this way, one reaches finitely many common zeros of $\mathfrak{a}$ lying in an algebraic extension field of $P(u; x_{i+1} \ldots x_n)$. Conversely, because $\mathfrak{a}_i$ is divisible by $\mathfrak{a}$, every zero of $\mathfrak{a}$ is also a zero of $\mathfrak{a}_i$. If, in particular, $x_1 = \bar{x}_1, \ldots, x_i = \bar{x}_i, x_{i+1} = x_{i+1}, \ldots, x_n = x_n$ is a zero of $\mathfrak{a}$, then $\mathfrak{a}_{i+1} = 0$ follows, and the polynomials from $\mathfrak{a}_i$ acquire a greatest common divisor with the linear factor $x_i - \bar{x}_i$. Summarizing:

Theorem XII. *Let $\mathfrak{a}$ be different from the unit ideal, with $\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_i \neq 0$ and $\mathfrak{a}_{i+1} = 0$. Then, after adjoining $x_{i+1} \ldots x_n$ to $P(u)$, there exist finitely many associated value systems $\bar{x}_1 \ldots \bar{x}_i$, lying in an algebraic extension field of $P(u, x_{i+1}, \ldots, x_n)$, such that all polynomials from $\mathfrak{a}$ vanish for $x_1 = \bar{x}_1, \ldots, x_i = \bar{x}_i, x_{i+1} = x_{i+1}, \ldots, x_n = x_n$. Conversely, if such a zero of $\mathfrak{a}$ is present, then $\mathfrak{a}_{i+1} = 0$ follows, as does the occurrence of a common divisor $D^{(i)}(x)$ of all polynomials from $\mathfrak{a}_i$ that has the linear factor $x_i - \bar{x}_i$.*

[4] This is in principle the same inference on which the isomorphism of $\mathfrak{G}^*_{i-1} \mid \mathfrak{M}^*_{i-1}$ with $\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}$ rested.

Theorem XII shows in particular that every ideal $\mathfrak{a}$ having no zero is the unit ideal.

To carry out a decomposition that also explicitly exhibits the association of the value systems, introduce new indeterminates v, which may be adjoined to $P(u, x_{i+1}, \dots, x_n)$. Put

$$z_i = -v_1 x_1 - \dots - v_{i-1} x_{i-1} + x_i. \tag{35}$$

Then the composition of (12) with (35) is again a substitution of type (12), in which now only the u_{ik} are replaced by $w_{ik} = u_{ik} - v_k$, and more generally $u_{i+h,k}$ by $w_{i+h,k} = u_{i+h,k} - u_{i+h,i} v_k$. If the substitution (35) carries the, by hypothesis transformed, ideal $\mathfrak{a}$ into $\mathfrak{b}$, then $\mathfrak{b}$ is obtained from $\mathfrak{a}$ simply by replacing the $u_{\mu\nu}$ by $w_{\mu\nu}$ and adjoining the v; and by the same process the ideals $\mathfrak{b}_1, \mathfrak{b}_2, \dots$ defined by $\mathfrak{b}$ arise from $\mathfrak{a}_1, \mathfrak{a}_2, \dots$. Conversely $\mathfrak{a}_i$ is obtained from $\mathfrak{b}_i$ by $v = 0$; therefore the ideals $\mathfrak{a}_i$ and $\mathfrak{b}_i$ are simultaneously zero ideals or simultaneously different from the zero ideal.

Now again let $\mathfrak{a}_1 \neq 0; \dots; \mathfrak{a}_i \neq 0; \mathfrak{a}_{i+1} = 0$, and hence also $\mathfrak{b}_1 \neq 0; \dots; \mathfrak{b}_i \neq 0; \mathfrak{b}_{i+1} = 0$. Let $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ be an associated zero-system of $\mathfrak{a}$. Define $\bar{z}_i = \bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1}$. Then $\bar{x}_1 \dots \bar{x}_{i-1}, \bar{z}_i, x_{i+1} \dots x_n$ is, by (35), a zero of $\mathfrak{b}$. Hence if $H^{(i)}(z, x)$ denotes the greatest common divisor of all polynomials from $\mathfrak{b}_i$, which may be assumed integral and primitive in the v, then by the last part of Theorem XII, $H^{(i)}(z, x)$ has the factor

$$z_i - \bar{z}_i = z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1}),$$

and to every zero of $\mathfrak{a}$ occurring after adjoining $x_{i+1} \dots x_n$ there corresponds such a factor. It must be shown that this exhausts the factorization of $H^{(i)}(z, x)$, i.e. that no factor $H_1^{(i)}$ can be separated from $H^{(i)}$ which does not decompose into linear factors in z and v. If this were the case, then because of the assumed primitivity in v, $H^{(i)}$ would contain at least one linear factor $z_i - \bar{z}_i$, where $\bar{z}_i$ belongs to an algebraic extension field of $P(u, v, x_{i+1}, \dots, x_n)$. If $\bar{x}_1 \dots \bar{x}_{i-1}, \bar{z}_i, x_{i+1} \dots x_n$ is the corresponding zero of $\mathfrak{b}$, then it must coincide with one of the finitely many zeros of $\mathfrak{a}$ that occur after adjoining $x_{i+1} \dots x_n$, and so it must be independent of v. Hence $\bar{z}_i$ necessarily contains the v linearly, not algebraically or rationally non-linearly; contrary to the assumption, another linear factor $z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1})$ in z and v can be split from $H_1^{(i)}$. Thus the explicit decomposition is

$$H^{(i)}(z, x) = \prod \{ z_i - (\bar{x}_i + v_1 \bar{x}_1 + \dots + v_{i-1} \bar{x}_{i-1}) \}^{\alpha_\lambda}.$$

Since the degree of $H^{(i)}$ and the individual exponents α_λ must agree with the corresponding ones for $D^{(i)}(x)$ – for $H^{(i)}$ arises from $D^{(i)}$ by the specialization $u_{\mu\nu} = w_{\mu\nu}$, and $D^{(i)}$ from $H^{(i)}$ by the specialization $v = 0$ – this decomposition also shows that, because the indeterminates $u_{\mu\nu}$ are adjoined to P, every zero $\bar{x}_i$ of $D^{(i)}$ occurring in the elimination starting from $\mathfrak{a}$ corresponds to a greatest common divisor from $\mathfrak{a}_{i-1} \dots \mathfrak{a}_1$ that is respectively linear, or to a power of such; hence in each case only one associated zero-system $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ of $\mathfrak{a}$ is obtained.[5]

§ 7. Resultant form and elimination theory.

To establish the connection between the resultant form and elimination theory, apply the successive elimination of § 6 specifically to the quotient $\mathfrak{a} = \mathfrak{m}/\mathfrak{g}_{i-1}$ of ideal by fundamental ideal of the $(i-1)$-st stage. It must first be shown that this quotient is a transformed ideal. Now $\mathfrak{m}$ is transformed and hence, by Theorem VI of § 3, so is $\mathfrak{g}_{i-1}$. From

$$H(x) \equiv 0(\mathfrak{m}/\mathfrak{g}_{i-1}); \qquad H(x) = \overline{H}(y) = \sum \alpha_i(u) \overline{H}_i(y) = \sum \alpha_i(u) H_i(x)$$

one obtains

$$\overline{H}(y) \overline{\mathfrak{g}}_{i-1,(u)} \equiv 0(\overline{\mathfrak{m}}_{(u)}), \quad \text{hence also} \quad \overline{H}(y) \overline{\mathfrak{g}}_{i-1} \equiv 0(\overline{\mathfrak{m}}_{(u)}),$$

or

$$\overline{H}_i(y) \overline{\mathfrak{g}}_{i-1} \equiv 0(\overline{\mathfrak{m}}), \quad \text{hence} \quad H_i(x) \mathfrak{g}_{i-1} \equiv 0(\mathfrak{m}).$$

This proves the divisibility of $H_i(x)$ by $\mathfrak{m}/\mathfrak{g}_{i-1}$ and hence that this ideal is transformed, so that the elimination method of § 6 applies.

[5]It should be pointed out that the successive elimination given here, in contrast to Kronecker's method, depends only on the given ideal $\mathfrak{a}$ and is independent of every ideal basis; in particular this also holds for the exponents α_λ. A complete execution of the elimination by this method would, according to Hentzelt, require the continuation of the procedure on the quotient $\mathfrak{a}/D^{(i)}$, which may be omitted in view of the next section. (E. N.)

But (30), taking account of (28) – § 4 – shows that $C^{(i)}(x)$ is divisible by $\mathfrak{m}/\mathfrak{g}_{i-1}$, where $C^{(i)}(x)$ denotes any non-identically vanishing determinant of degree ρ from $\mathfrak{M}^*_{i-1}$. Since for $i>1$ this $C^{(i)}$ has degree zero in x_1, the module $\mathfrak{A}_1$ corresponding to the ideal $\mathfrak{a}=\mathfrak{a}_1$ contains the polynomials $C^{(i)}, x_1C^{(i)}, \dots, x_1^{k_1-1}C^{(i)}$, and therefore $(C^{(i)})^{k_1}$ is divisible by $\mathfrak{a}_2$; similarly $(C^{(i)})^{k_1k_2}$ is divisible by $\mathfrak{a}_3$, and finally $(C^{(i)})^{k_1k_2\cdots k_{i-1}}$ by $\mathfrak{a}_i$. Thus $\mathfrak{a}_1,\dots,\mathfrak{a}_i$ are different from the zero ideal, as also holds for $i=1$. Let $D^{(i)}(x)$ now be the greatest common divisor of all polynomials from $\mathfrak{a}_i$, which has degree $p_i>0$ only when $\mathfrak{a}_{i+1}$ is the zero ideal. By the definition of $D^{(i)}(x)$, and taking into account the divisibility of $\mathfrak{a}_i$ by $\mathfrak{a}$, one obtains

$$d^{(i+1)}D^{(i)}(x) \equiv 0(\mathfrak{m}/\mathfrak{g}_{i-1}); \qquad d^{(i+1)} \neq 0.$$

Thus $d^{(i+1)}D^{(i)}$ is a polynomial, free of $x_1 \dots x_{i-1}$, belonging to $\mathfrak{m}/\mathfrak{g}_{i-1}$. But $E^{(i)}(x)$ was defined (Theorem II and § 4), as the highest elementary divisor of $\mathfrak{M}_{i-1}$ or $\mathfrak{M}^*_{i-1}$, to be the greatest common divisor, in the polynomial sense, of all polynomials from $\mathfrak{m}/\mathfrak{g}_{i-1}$ that are free of $x_1 \dots x_{i-1}$. Therefore $d^{(i+1)}D^{(i)}$, and because of the regularity of $E^{(i)}(x)$ in x_i also $D^{(i)}(x)$, is divisible by $E^{(i)}(x)$.

The same argument can be carried out if the transformation (12) had from the beginning been composed with (35), that is, if the indeterminates $u_{\mu\nu}$ had been replaced by $w_{\mu\nu}$; this gives the divisibility of $H^{(i)}(z,x)$ by the corresponding $\bar{E}^{(i)}(z,x)$. Since, just as $D^{(i)}(x)$ and $H^{(i)}(z,x)$ pass into one another by specialization, so too do $E^{(i)}(x)$ and $\bar{E}^{(i)}(z,x)$, and also $R^{(i)}(x)$ and $\bar{R}^{(i)}(z,x)$, the multiplicity numbers in the decomposition into linear factors must also agree mutually. Taking into account, furthermore, that a power of $E^{(i)}(x)$ is divisible by $R^{(i)}(x)$, one obtains the following summary.

Theorem XIII. *If $R^{(i)}(x)$ is decomposed in an algebraic extension field of $P(u, x_{i+1}, \dots, x_n)$ into linear factors $x_i - \bar{x}_i$, then $\bar{x}_i$ can be completed in one and only one way to an associated value system $\bar{x}_1 \dots \bar{x}_i, x_{i+1} \dots x_n$ that is a zero of $\mathfrak{m}/\mathfrak{g}_{i-1}$ and consequently of $\mathfrak{m}$. The finitely many zeros of $\mathfrak{m}$ arising in this way from the linear factors of $R^{(i)}$ – finitely many after adjoining $x_{i+1} \dots x_n$ – are collected by the explicit decomposition*

$$\bar{R}^{(i)}(z,x) = \prod \{z_i - (\bar{x}_i + v_1\bar{x}_1 + \cdots + v_{i-1}\bar{x}_{i-1})\}^{\alpha_\lambda}. \tag{36}$$

Because of the regularity of $\bar{R}^{(i)}(z,x)$ in z_i, this decomposition is preserved if the $x_{i+1} \dots x_n$ adjoined to $P(u)$ are replaced by arbitrary special value systems $\bar{x}_{i+1} \dots \bar{x}_n$ from $P(u)$ or from the algebraically closed field derived from it.

This also accomplishes a separation of the zeros of $\mathfrak{m}$ according to the individual factors of the resultant form; the number of indeterminates x adjoined to $P(u)$ is also called the dimension of the corresponding algebraic object.

If, in the decomposition of $\bar{R}^{(i)}(z,x)$ or of the corresponding $R^{(i)}(x)$, the factors conjugate over $P(u, x_{i+1} \dots x_n)$ are collected together – factors that for general P may be wholly or partially identical – one obtains a decomposition of $R^{(i)}(x)$ into powers of polynomials irreducible over $P(u, x_{i+1} \dots x_n)$:

$$R^{(i)}(x) = S_1^{(i)}(x)^{\beta_1} \cdots S_\nu^{(i)}(x)^{\beta_\nu}.$$

By Theorem X, this decomposition corresponds to a representation $\mathfrak{g}_i = [\mathfrak{j}_1 \dots \mathfrak{j}_\nu]$ such that $S_\mu^{(i)}(x)^{\beta_\mu}$ is equal to the norm of $\mathfrak{g}_{i-1}$ with respect to the ideal $\mathfrak{j}_\mu$, respectively to the norm of the module $\mathfrak{G}_{i-1}$ with respect to the module corresponding to $\mathfrak{j}_\mu$. This clarifies the meaning of the exponents β_μ: in particular, if γ_μ is the degree of $S_\mu^{(i)}$, then the multiplicity $\beta_\mu\gamma_\mu$ is equal to the number of residue classes of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{j}_\mu$ that are linearly independent over $P(u, x_{i+1} \dots x_n)$.

Finally consider briefly the special case where P has characteristic zero. If the $u_{\mu\nu}$ are specialized to quantities $\bar{u}_{\mu\nu}$ from P in such a way that the n determinants $C^{(i)}(x)$ $(i=1,2,\dots,n)$ remain regular in x_i,[6] then the regularity of $R^{(i)}$ is also preserved. Under this specialization, $R^{(i)}$ and likewise $\bar{R}^{(i)}(z,x)$ become a common divisor of all ρ-rowed determinants from $\mathfrak{M}^*_{i-1}$, though not necessarily the greatest common divisor. But the specialization can always be chosen so that this latter property also remains true. For, as the existence of the module basis shows, $R^{(i)}(z,x)$ can also be characterized as the greatest common divisor of finitely many ρ-rowed determinants from $\mathfrak{M}^*_{i-1}$. The decomposition of the so specialized $\bar{R}^{(i)}(z,x)$ is then obtained simply by replacing the $u_{\mu\nu}$ by $\bar{u}_{\mu\nu}$ in (36), so that the whole elimination theory is preserved.

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[6] Hentzelt assumes this specialization of the $u_{\mu\nu}$ from the start and therefore has to call upon much more complicated considerations to prove the decomposability of $H^{(i)}(z,x)$ and $\bar{R}^{(i)}(z,x)$ into linear factors in z and v. The exponents that occur there need not necessarily agree with the ones above. (E. N.)